

Employee Attrition Analysis, ABC Manufacturing Ltd

AnalystLab Africa | Data Science Internship Programme, Week 1
Project: Business Understanding & Data Exploration

Overview

This project investigates employee attrition at ABC Manufacturing Ltd on behalf of AnalystLab Africa Consulting. The goal of this phase is to understand the business problem, explore the workforce dataset, and generate evidence-based insights before any predictive modelling is considered.

Dataset

[IBM HR Analytics – Employee Attrition & Performance](#) (Kaggle), 1,470 employee records, 35 features.

Business Questions

1. What does the company's workforce look like?
2. Which departments have the highest employee attrition?
3. Does age influence attrition?
4. Does monthly income affect retention?
5. Does overtime influence attrition?
6. Which job roles experience the highest turnover?
7. Which variables appear important for future predictive modelling?

Repository Contents

File	Description
attrition_analysis.ipynb	Full Jupyter notebook, data loading, inspection, and visualisations
Business_Understanding_Report.docx	Business context, impact of attrition, and rationale for this engagement
Dataset_Inspection_Report.docx	Data structure, quality checks, and descriptive statistics
Data_Visualisation_Report.docx	Charts and interpretations (bar, histogram, boxplot, pie)
Business_Insights_Report.docx	Key workforce observations and attrition risk factors
Reflection_Report.docx	Personal reflection on the week's learning
WA_Fn-UseC_-HR-Employee-Attrition.csv	Source dataset

Key Findings

- **Overtime** is one of the strongest attrition drivers: 30.5% attrition among employees who work overtime vs. 10.4% among those who don't.
- **Income** matters: the lowest income quartile has a 29.3% attrition rate vs. 10.3% in the highest quartile.
- **Age** matters: employees aged 18–25 leave at 34.8%, compared to just 9.2% for the 36–45 bracket.
- **Job role** matters: Sales Representatives have the highest attrition rate (39.8%) of any role.
- **Department**: Sales has the highest departmental attrition rate (20.6%), ahead of HR (19.0%) and R&D (13.8%).

Tools Used

Python, Pandas, Matplotlib, Seaborn, Google Colab

Author

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Week 2: Feature Engineering & Data Preprocessing for Machine Learning

Building on Week 1, this phase cleans, transforms, encodes, scales, and engineers features to produce a machine-learning-ready dataset for Week 3 model development.

Additional Files

File	Description
<code>Week2_Feature_Engineering.ipynb</code>	Full preprocessing notebook (cleaning → feature engineering → encoding → scaling)
<code>Week2_Business_Understanding_Report.docx</code>	Why preprocessing matters for this engagement
<code>Data_Preprocessing_Report.docx</code>	Full documentation of the preprocessing pipeline
<code>cleaned_dataset.csv</code>	Cleaned dataset (1470 × 31), constant/ID columns removed, still human-readable
<code>ml_ready_dataset.csv</code>	Machine-learning-ready dataset (1470 × 53), encoded and scaled

Pipeline Summary

Stage	Shape	Notes
Raw dataset	1470 × 35	As downloaded from Kaggle
Cleaned dataset	1470 × 31	Dropped 3 constant columns + 1 ID column
Feature-engineered	1470 × 36	Added 5 new derived features
ML-ready (final)	1470 × 53	One-hot encoding expanded categoricals; numeric features standardised

Engineered Features

- **AverageSatisfaction**, composite of 4 related satisfaction scores
- **TenureRatio**, $\text{YearsAtCompany} / \text{TotalWorkingYears}$ (removed in Week 3, see below)
- **IncomePerExperienceYear**, $\text{MonthlyIncome} / \text{TotalWorkingYears}$
- **AgeGroup**, age bracket matching Week 1's EDA (18-25, 26-35, 36-45, 46-55, 56-60)
- **PromotionStagnationRatio**, $\text{YearsSinceLastPromotion} / \text{YearsAtCompany}$ (removed in Week 3, see below)

Week 3: Advanced Data Analysis, Statistical Validation & Feature Engineering

Building on the cleaned Week 2 dataset, this phase conducts deeper exploratory analysis, formal statistical hypothesis testing, additional feature engineering, and feature evaluation, producing the Final Modelling Dataset for Week 4 machine learning development.

Additional Files

File	Description
Week3_Advanced_EDA.ipynb	Advanced EDA notebook, 13 visualisations covering correlation, multivariate, and target-variable analysis
Project_Continuity_Summary.docx	Recap connecting Week 1 to Week 3, including open questions carried forward
Statistical_Analysis_Summary.docx	4 formal hypothesis tests (Chi-Square, 2x Mann-Whitney U, Kruskal-Wallis) with full interpretation
Feature_Engineering_Documentation.docx	4 new engineered features with justification and evidence
Feature_Evaluation_Summary.docx	Correlation/redundancy analysis and retain/remove decisions for all 9 engineered features
Business_Insights_Recommendations_Report.docx	Executive summary, findings, recommendations, limitations, next steps
Data_Dictionary.docx	Full column-by-column reference for the Final Modelling Dataset
Final_Modelling_Dataset.csv	Final ML-ready dataset (1470 × 54), Week 4-

File	Description
	ready
requirements.txt	Python package dependencies for this project

Key Statistical Findings

- **Department x OverTime:** not significant (Chi-Square, $p = 0.9543$), overtime is evenly spread across departments, ruling it out as the explanation for Sales' higher attrition.
- **Income by Attrition:** significant (Mann-Whitney U, $p < 0.0001$), median income 3,202 (left) vs 5,204 (stayed).
- **Age by Attrition:** significant (Mann-Whitney U, $p < 0.0001$), median age 32 (left) vs 36 (stayed).
- **Income across Job Role:** significant (Kruskal-Wallis, $p < 0.0001$), Sales Representative has the lowest median income (2,579) of any role in the company, plausibly explaining its highest attrition rate (39.8%, from Week 1).

New Engineered Features (Week 3)

- **HighRiskProfile**, binary flag for employees who are simultaneously young (≤ 35), lower-paid (below median), and less experienced (≤ 10 years); 27.0% attrition rate vs 11.4% otherwise
- **LogMonthlyIncome**, log transform of MonthlyIncome, reduces skewness from 1.37 to 0.29
- **JobHopRate**, $\text{NumCompaniesWorked} / (\text{TotalWorkingYears} + 1)$
- **IncomeBand**, Low/Medium/High/Very High quartile bins of MonthlyIncome (reporting only, excluded from modelling dataset)

Feature Evaluation Outcome

Two Week 2 engineered features were removed from the Final Modelling Dataset based on statistical evidence of negligible relationship with Attrition:

- **TenureRatio** ($r = -0.016$ with Attrition)

- **PromotionStagnationRatio** ($r = 0.03$ with Attrition; contradicted by Week 3 EDA)

Pipeline Summary (Updated)

Stage	Shape	Notes
Raw dataset	1470 x 35	As downloaded from Kaggle
Cleaned dataset	1470 x 31	Dropped 3 constant columns + 1 ID column
Week 2 feature-engineered	1470 x 36	Added 5 new derived features
Week 3 feature-engineered	1470 x 40	Added 4 more derived features
Final Modelling Dataset	1470 x 54	2 low-value Week 2 features removed; remaining features encoded and scaled